

RESEARCH ARTICLE

**Economic Disadvantage and Mass Shootings
Across U.S. States, 2018 to 2024:
A State-Level Panel Analysis**

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Abstract

Background: We asked whether poverty, gun-law strength, or serious mental illness were associated with higher mass-shooting rates across U.S. states.

Methods: We counted mass shootings (four or more people shot, not counting the shooter) in the 50 states, 2018 to 2024, from the Gun Violence Archive (3,783 incidents); only 5% had four or more killed. We compared state rates per million residents with negative-binomial regression, accounting for the year, and checked how strong an unmeasured factor would need to be to erase each finding.

Results: Mass shootings roughly doubled after 2019 and stayed high. Each percentage point of poverty was associated with about 15% more mass shootings per million residents (95% CI 6% to 26% more). The 10 highest-poverty states had 1.7 times the rate of the 10 lowest, though the rise was not steady. Poverty held up best against unmeasured factors and stayed linked when we added violent crime or the share of Black residents. That share, which we read as a sign of segregation and concentrated disadvantage, was itself strongly linked and halved the poverty link; an unmeasured factor as strong as that share, the strongest factor we measured, could erase it. Poverty could not be separated from low income in analyses giving large states more weight, and other methods disagreed on which economic measure mattered most. Gun-law strength showed no clear link (4% more per full letter grade; 95% CI 10% fewer to 20% more). Within regions, stronger-law states had 19% more, but this rested mostly on Illinois. States with more serious mental illness had 18% fewer per percentage point, but this was fragile, disappearing with the share of Black residents added. Twenty cities or counties held 32% of incidents.

Conclusions: Where poverty and disadvantage concentrate, mass shootings concentrate too. Prevention resources belong in those places, and economic investment there is worth testing, alongside gun policy. State data cannot settle whether gun laws work; that needs local data. State mental-illness rates are not a guide to where shootings happen. Results may differ for fatal or public mass shootings. These are associations, not proof of cause.

Keywords: mass shootings, poverty, concentrated disadvantage, gun policy, mental health, violence prevention, health policy

Background

Mass shootings take lives, injure many more people, and hurt the families and neighborhoods around them. Knowing where they happen, and what those places have in common, tells communities and policymakers where prevention can do the most good.

In the United States, their number rose sharply after 2019 [1]. Public debate about prevention tends to focus on two levers, gun laws and mental illness. Much less attention goes to the economic conditions of the places where mass shootings happen.

The research that does exist points to economic conditions. Kwon and Cabrera [2] and Cabrera and Kwon [3] found that U.S. counties and states with greater income inequality had higher mass-shooting rates, and read this through Merton’s strain theory, in which the gap between visible affluence and blocked opportunity creates conditions for violence [4]. Income inequality and poverty move together, however, and few studies compare economic conditions, gun policy, and mental health side by side.

Earlier state-level studies of gun laws disagree, and the disagreement follows the definition of a mass shooting. For shootings with four or more people killed, from 1998 to 2015, states with more permissive gun laws and higher gun ownership had higher rates [5]. For the broader definition of four or more people shot, which we also use, rates from 2014 to 2021 did not differ by state gun-law grade [6]. Machine-learning methods have also been used to look for the conditions that predict firearm violence in cities [7]. This paper adds four things: seven years of data on all 50 states; economic conditions, gun-law strength, and mental illness in one model; a check of how strong an unmeasured factor would have to be to change each finding, compared with the factors we did measure; and a full report of the analyses whose results disagreed with the main ones.

We asked three questions. (1) How did mass shootings change from 2018 to 2024? (2) Were poverty, gun-law strength, and serious mental illness linked to state mass-shooting rates? (3) Did the poverty finding hold up under simple checks?

Methods

Study design and outcome

This is an observational study that compares states, not people. Each of the 50 states is observed once in each year from 2018 to 2024. Mass-shooting counts came from the Gun Violence Archive (GVA), which defines a mass shooting as an incident in which four or more people are shot (injured or killed), not counting the shooter [1]. This definition captures the broad burden of community gun violence. It differs from fatal-only definitions (four or more

killed), which pick out a rarer and different set of events [8]. Most incidents counted here were not mass killings: of the 3,783 incidents in the 50 states, 5% had four or more people killed and in 46% no one was killed (victims only, not counting the shooter). Incident records were taken from an automatically updated, incident-level copy of GVA’s public mass-shooting reports maintained on GitHub (repository dxzys/Gun-Violence-Data, commit 3d2c2b8 of 6 October 2026) [9] and counted by unique incident ID for each state and calendar year; no incidents were excluded. The repository carries no license, and GVA states that its data are copyrighted, so we used the copy only to count incidents and share only the derived state-year counts. As a check against the source, the national totals in this copy (all incidents, including the District of Columbia) were within 3 incidents of GVA’s own published annual totals in every year from 2018 to 2024 [1]. The state-year counts were also stable across two retrievals of the copy eight months apart (10 February and 6 October 2026). Because the GVA is continuously updated, counts may differ slightly by retrieval date.

The data hold 350 state-years (50 states $\times$ 7 years). Each state characteristic was measured once, so comparisons between states rest on 50 states, observed seven times. The District of Columbia was excluded: it is not a state and has no Giffords gun-law grade.

State characteristics

An earlier version of this manuscript used a compiled dataset whose values could not all be traced to their stated sources. For this version, every state characteristic was extracted again, directly from its official source file, by a single script. Additional file 2 gives the source, table, field, and web address of every variable, and Additional file 4, Tables S1 and S2, summarize those used here. All estimates reported here replace the earlier ones. No value is missing.

The three characteristics in the main model were the poverty rate (percent of people below the poverty level, 2023 American Community Survey [ACS] [10]); gun-law strength (the Giffords Law Center 2023 letter grade [11], converted to a 0 to 4 scale: A 4.0, A– 3.7, B+ 3.3, B 3.0, B– 2.7, C+ 2.3, C 2.0, C– 1.7, D+ 1.3, D 1.0, D– 0.7, F 0); and the percent of adults with serious mental illness in the past year (2022 to 2023 National Survey on Drug Use and Health [NSDUH] state estimates [12]). Poverty is the main model’s economic measure because the ACS poverty rate is the standard official measure of economic hardship for U.S. states and the one most directly tied to concentrated disadvantage; median household income and income inequality are reported alongside it throughout. The checks used median household income and the percent of residents who are Black (both ACS 2023 [10]; percent Black is built by a separate script into its own file, Additional file 3). Income

inequality was measured by the Gini index (ACS 2023). Rates per million residents used the July 1, 2023 state population (Census Population Estimates Program, Vintage 2024 [13]), and Census regions came from the same file. Other analyses also used household gun ownership (2016, the latest year in the RAND state estimates [14]), violent crime (2023 FBI estimates [15]), percent urban (2020 Census [16]), and the other state characteristics listed in Additional file 2.

Analysis

The main model. We compared mass-shooting rates per million residents across states with a count model, a regression built for counts of events that also adjusts for each state’s population. All three characteristics were in the model together. The model accounted for the year, so that the national rise after 2019 is not mistaken for differences between states, and for the fact that the seven years from the same state are related. In technical terms, this is negative-binomial regression with the logarithm of state population as an offset, calendar-year fixed effects, and cluster-robust standard errors by state with the CR2 small-sample correction and Satterthwaite degrees of freedom [17,18]. We used the negative binomial rather than the Poisson model because the counts varied far more than a Poisson model allows (Pearson dispersion statistic 4.3 for the main model’s terms, against 1 expected under the Poisson model).

Reading the results. We report each characteristic as an incidence rate ratio (IRR) and as the percent change in the mass-shooting rate for one natural unit: one percentage point of poverty, one full letter grade of gun-law strength (one point on the 0 to 4 scale, for example from C to B), and one percentage point of adults with serious mental illness. Each comes with a 95% confidence interval (CI): the range of values consistent with the data. When the 95% CI includes no change, the data do not show a clear association. These are associations across states, each with the other two characteristics and the year held equal.

An observed comparison. As a check that needs no model, we compared the observed mass-shooting rate per million residents per year in the 10 states with the highest poverty rates and in the 10 with the lowest, along with each group’s share of the population and of the incidents. To see whether that contrast reflects a steady pattern, we also split the 50 states into five groups of 10, from lowest to highest poverty, and plotted each state’s rate (Figure 1). We also counted incidents by city or county, as recorded by GVA, to see where they concentrate.

Three checks of the poverty finding. We refit the main model (a) 50 times, leaving out one state each time; (b) comparing each state only with states in its own Census region

(Northeast, Midwest, South, West); and (c) adding median household income, and separately adding the percent of residents who are Black. The percent Black is a check, not part of the main model, because at the state level it is a proxy for the same thing this paper is about, concentrated disadvantage: adjusting for it partly removes the exposure rather than a confounder, so we report it as a bound on how much of the poverty association is shared with segregation.

How strong would a hidden factor have to be? Because this study is observational, a factor we did not measure could go with both a state characteristic and its mass-shooting rate. We used the sensitivity method of Cinelli and Hazlett [19] to ask how strong such a hidden factor would have to be to erase each of the three findings, and compared that with the strongest factors we did measure: the share of Black residents, median household income, and violent crime. The method needs a simple linear model, so it uses one row per state: the log of each state’s 2018 to 2024 mass-shooting rate, with poverty, gun-law strength, and serious mental illness, fit by ordinary least squares with conventional standard errors, as in the standard form of the method. This linear model weights every state equally, whereas the count model with a population offset gives large states more weight, so the two answer slightly different questions. To see whether that difference matters, we refit the linear model with each state weighted by its population (weighted least squares) and repeated the analysis for poverty.

Other analyses. We also ran other versions of the main model (for example, with income inequality or median household income in place of poverty, with household gun ownership and violent crime added, or with the years before and after 2020 analyzed separately), looked at each characteristic on its own, and used three methods that rank state characteristics by how well they predict each state’s mass-shooting rate: LASSO [20], random forests [21], and gradient boosting [22]. With only 50 states, the ranking methods are a rough check, not a test. Additional file 4 describes each in plain words and gives every result, including those that point another way. In the version with one row per state (Additional file 4, Table S4), each state is its own cluster, so the CR2 correction reduces to the HC2 heteroskedasticity-robust standard error, again with Satterthwaite degrees of freedom.

Analyses were run in R 4.4.0 (packages `MASS` and `clubSandwich` for the main model). One script fits every model and writes every reported number, so all results can be reproduced from the shared data and code.

Use of AI-assisted technologies

Generative AI (Claude, Anthropic) was used to assist with writing, running and checking the analysis code and with drafting, editing and proofreading the text. The author designed the study, made all analytic and interpretive decisions, reviewed all code and results, verified the reported numbers and references, and takes full responsibility for the content. No AI tool meets authorship criteria.

Results

Mass shootings roughly doubled after 2019 and stayed high

The 50 states recorded 3,783 mass shootings from 2018 to 2024. National counts rose from 329 in 2018 to 673 in 2021, 2.0 times as many, stayed near that level through 2023, and fell to 491 in 2024, still 1.5 times the 2018 count (Table 1). In the main model, with the state characteristics held equal, the incidence rate ratio for every year from 2020 to 2023 compared with 2018 was 1.89 to 2.01. The surge was the largest signal in the data. Because the state characteristics were each measured once, this study documents the surge but cannot explain it.

Table 1: Mass shootings by year in the 50 states, and how each year compares with 2018

Year	Incidents	IRR compared with 2018 (95% CI)	<i>p</i>
2018	329	1 (reference)	
2019	406	1.25 (1.03 to 1.52)	.027
2020	602	1.89 (1.59 to 2.25)	< .001
2021	673	2.01 (1.65 to 2.46)	< .001
2022	633	1.91 (1.56 to 2.34)	< .001
2023	649	1.98 (1.62 to 2.41)	< .001
2024	491	1.56 (1.29 to 1.89)	< .001
Total	3,783		

Incidents: Gun Violence Archive mass shootings (four or more people shot, not counting the shooter) in the 50 states. IRR: incidence rate ratio from the main model, the year's mass-shooting rate per million residents divided by the 2018 rate, with state poverty, gun-law strength, and serious mental illness held equal; an IRR of 2 means twice the 2018 rate. CI: confidence interval.

Where mass shootings happened

Incidents were concentrated in a small number of places. They occurred in 1,114 cities or counties, but the 20 places with the most incidents accounted for 1,217 of them (32%). In the typical state with at least 10 incidents (the median of 37 states), a single city or county accounted for 43% of the state's incidents; in Illinois, Chicago accounted for 74%.

Higher-poverty states had more mass shootings

The plainest comparison needs no model. The 10 states with the highest poverty rates (14.0% to 18.9% of residents below the poverty level) had 2.2 mass shootings per million residents per year, 1.7 times the rate of 1.3 in the 10 states with the lowest poverty rates (7.2% to 10.2%). The high-poverty states held 16% of the population but 22% of the incidents; the low-poverty states held 11% of the population and 9% of the incidents.

The rise across poverty levels was not steady, however (Figure 1). With the 50 states split into five groups of 10, from lowest to highest poverty, the rates were 1.3, 1.0, 1.9, 1.5, and 2.2 mass shootings per million residents per year. The two lowest-poverty groups had the lowest rates and the highest-poverty group had the highest, but the middle group, which includes Illinois, had a higher rate than the group above it. States within each group also differed widely (Figure 1).

The main model shows the same overall pattern (Table 2). With gun-law strength, serious mental illness, and the year held equal, each percentage point of poverty was associated with an incidence rate ratio of 1.15 (95% CI 1.06 to 1.26; $p = .003$), that is, about 15% more (95% CI 6% to 26% more) mass shootings per million residents.

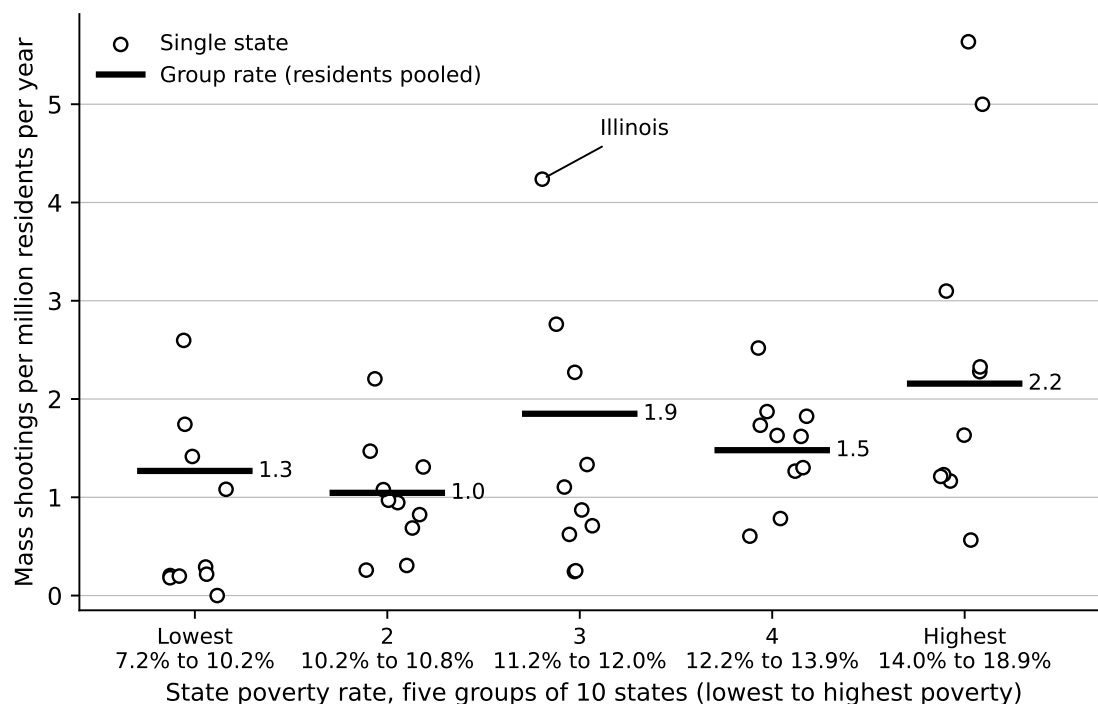


Figure 1: Mass-shooting rates by state poverty level, 2018 to 2024. Each circle is one state: its 2018 to 2024 mass shootings divided by 7 and by its July 1, 2023 population, per million residents. The 50 states are split into five groups of 10 by poverty rate (2023 American Community Survey), from lowest to highest; the range of poverty rates in each group is shown below it. The bar and the number beside it give the group rate, with all residents of the 10 states pooled, so large states count more in the bar than in the circles. Circles are spread sideways within each group only so that they do not overlap. Illinois, the state with the most incidents, is labeled. These are observed rates, not adjusted for any other characteristic.

Table 2: Main results: state characteristics and the mass-shooting rate per million residents

State characteristic, per unit	Range across the 50 states	IRR per unit (95% CI)	Change in rate, % (95% CI)	<i>p</i>
Poverty rate, per percentage point	7.2% to 18.9%	1.15 (1.06 to 1.26)	15 (6 to 26)	.003
Gun-law strength, per full letter grade (one point on the 0 to 4 scale)	0 (F) to 4 (A)	1.04 (0.90 to 1.20)	4 (−10 to 20)	.595
Serious mental illness, per percentage point of adults	4.64% to 9.34%	0.82 (0.69 to 0.98)	−18 (−31 to −2)	.032

All three characteristics are in one model that also accounts for the year ($n = 350$ state-years, 50 states). IRR: incidence rate ratio, the mass-shooting rate per million residents with one more unit of the characteristic divided by the rate at the original value (one unit lower), the other characteristics and the year held equal. Change in rate: (IRR minus 1) times 100. CI: confidence interval; when it includes no change (IRR 1), the data do not show a clear association. Model: negative-binomial regression with a population offset, calendar-year fixed effects, and CR2 cluster-robust standard errors by state with Satterthwaite degrees of freedom.

The poverty finding under three checks

Table 3 shows three checks. (a) Leaving out any one state did not remove the association, including Illinois, the state with the most incidents; the largest p value across the 50 refits was .010. (b) Comparing each state only with states in its own Census region gave nearly the same estimate. (c) Accounting for income and for racial composition did weaken it, as described next. The association was also similar before and after the surge: 14% more per percentage point in 2018 to 2019 and 16% more in 2020 to 2024 (Additional file 4, Table S4).

Table 3: Three checks of the poverty finding

Check	IRR per percentage point of poverty (95% CI)	Change in rate, % (95% CI)	<i>p</i>
Main model (for comparison)	1.15 (1.06 to 1.26)	15 (6 to 26)	.003
(a) Leaving out one state at a time (50 refits)	1.13 to 1.18	13 to 18	largest .010
Without Illinois	1.15 (1.05 to 1.25)	15 (5 to 25)	.005
(b) Comparing states within the same Census region	1.16 (1.06 to 1.26)	16 (6 to 26)	.003
(c) Accounting for median household income	1.08 (0.94 to 1.24)	8 (−6 to 24)	.241
Accounting for the share of Black residents	1.08 (1.01 to 1.15)	8 (1 to 15)	.026

Each row refits the main model (poverty, gun-law strength, serious mental illness, and year) with the change shown. For leaving out one state, the row gives the lowest and highest of the 50 estimates and the largest of the 50 *p* values. IRR: incidence rate ratio; CI: confidence interval. Standard errors as in Table 2.

Poverty overlaps with low income, inequality, and racial composition

Across states, poverty, low income, and racial composition go together, and these data cannot separate them. States with more poverty almost always have lower median incomes (correlation $r = -0.80$). On its own, each \$10,000 of median household income was associated with 20% fewer mass shootings per million residents (95% CI 4% to 34% fewer). With poverty and income in the same model, neither was clearly associated: poverty was associated with 8% more per percentage point (95% CI 6% fewer to 24% more; $p = .241$) and income with 21% fewer per \$10,000 (95% CI 42% fewer to 8% more; $p = .129$). The data cannot tell the two apart, so poverty here stands for economic disadvantage broadly, not poverty as distinct from low income. Income inequality behaved in a similar way. On its own, each 0.01 rise in the Gini index was associated with 23% more mass shootings (95% CI 5% to 43% more; $p = .015$), but with poverty in the model it was no longer clearly associated (9% more; 95% CI 7% fewer to 28% more; $p = .232$), while poverty stayed associated.

Other ways of analyzing the data, including three methods that rank state characteristics by how well they predict mass-shooting rates, gave mixed results for which economic measure matters most: poverty, income, and inequality were each ranked high by some methods and low by others (Additional file 4, Table S3).

The share of Black residents was itself strongly associated with mass-shooting rates. With

poverty, gun-law strength, and serious mental illness held equal, each 10 percentage points more Black residents was associated with 57% more mass shootings per million residents (95% CI 35% to 83% more; $p < .001$). That share is also higher, on average, in higher-poverty states (correlation $r = 0.45$), and with it in the model the poverty association was about half as large, 8% more (95% CI 1% to 15% more) per percentage point, but still present. This result needs care. Race does not cause violence, and this is not a finding about Black residents or about any person. At the state level, the share of Black residents marks residential segregation and the concentrated disadvantage that came with it, which in U.S. cities was shaped by historical, state-sanctioned discrimination in lending and the disinvestment that followed [23,24]. Part of what the main model attributes to poverty is shared with those conditions, and these data cannot separate them.

Gun-law strength showed no clear association

In the main model, gun-law strength was not clearly associated with mass-shooting rates: 4% more (95% CI 10% fewer to 20% more) per full letter grade ($p = .595$). The CI is wide, so the data are consistent with somewhat fewer mass shootings in stronger-law states, no difference, or more. When states were compared only with others in their own Census region, stronger-law states had more mass shootings: 19% more per full letter grade (95% CI 2% to 38% more; $p = .025$). This rested on a few states. It was no longer clear when any one of Colorado, Illinois, Maryland, and Texas was left out, and Illinois mattered most: without Illinois the estimate was 14% more per full letter grade (95% CI 1% fewer to 31% more; $p = .061$). The study therefore cannot tell whether state gun laws work, in either direction.

Household gun ownership was not associated with mass-shooting rates on its own: 9% more per 10 percentage points of households with a gun (95% CI 9% fewer to 29% more; $p = .332$). It was associated only in a larger model that also held gun-law strength equal (24% more; 95% CI 4% to 46% more; $p = .017$). Gun ownership and gun-law strength track each other closely (correlation $r = -0.77$): states with stronger laws have fewer gun-owning households. A model cannot reliably split the roles of two measures that move together this closely, so we do not read this as a separate finding.

More mental illness did not mean more mass shootings

States with a higher share of adults with serious mental illness did not have more mass shootings. In the main model they had fewer: 18% fewer (95% CI 2% to 31% fewer) per percentage point ($p = .032$). The same direction appeared in several other versions of the model and for any mental illness (Additional file 4, Tables S4 and S5). It was not a reliable

pattern, however. It disappeared when states were compared within their own region (6% fewer; $p = .468$) and when the share of Black residents was accounted for (3% more; $p = .739$). It is a pattern across states and says nothing about people with mental illness.

How strong would a hidden factor have to be?

Table 4 asks how strong a factor we did not measure would have to be to erase each finding, and checks what happened when the strongest factors we did measure were added. Poverty held up. To erase its association, a hidden factor would have to explain 47% of the leftover differences between states, in both poverty and mass-shooting rates, that the model does not already explain (30% to make the association no longer clear). For comparison, the strongest factor we measured, the share of Black residents, explains 38% of the leftover differences in rates and 18% of those in poverty; median household income explains 62% of the leftover differences in poverty but <1% of those in rates; violent crime explains 11% and 16%. With each of the three actually added, the 95% CI for poverty still excluded zero (Table 4). There are limits. A hidden factor as strong as the share of Black residents, on top of it, could erase the poverty association, and one as strong as violent crime would leave it borderline. Gun-law strength had nothing to erase: it was not clearly associated to begin with and did not become so when any measured factor was added, so state data cannot judge gun laws. The mental-illness pattern was fragile: a hidden factor explaining only 4% of the leftover differences would make it unclear, and adding the share of Black residents erased it.

The linear model behind this check weights every state equally, whereas the main count model gives large states more weight, and this matters for income. In the equally weighted linear model poverty stayed associated with income added, but weighted by population, as in the main model, it could not be separated from income, which remained associated with lower rates (figures in the note to Table 4). In the weighted model the five most populous states (California, Texas, Florida, New York, and Pennsylvania) hold 37% of the weight, so the two versions ask different questions, one about the typical state and one about where most residents live, rather than giving one right answer and one wrong one.

Table 4: How strong would a hidden factor have to be to erase each finding?

Finding	How strong a hidden factor must be to erase it	95% CI still excludes zero with share of Black residents / income / violent crime added?	Verdict
More mass shootings in higher-poverty states	Explain 47% of the leftover differences (30% to make it unclear)	Yes / Yes / Yes	Robust in the equally weighted linear check; in the main model and with states weighted by population, not separable from income
Same finding, states weighted by population	Explain 34% (11% to make it unclear)	No / No / Yes	Not separable from income; borderline with share of Black residents
Gun-law strength	Nothing to erase: not clearly associated to begin with	No / No / No	Inconclusive
Fewer mass shootings in states with more serious mental illness	Explain 29%, but only 4% to make it unclear	No / Yes / Yes	Fragile; do not interpret

Leftover differences: the differences between states, in both the characteristic and the mass-shooting rate, that the other two characteristics do not explain. The strongest measured factor, the share of Black residents, explains 38% of the leftover differences in rates and 18% of those in poverty (33% and 7% with states weighted by population). Poverty and income: equally weighted, poverty with income added was associated with 18% more mass shootings per percentage point (95% CI 3% to 36% more; $p = .020$). Weighted by population, poverty alone was associated with 13% more per percentage point (95% CI 4% to 23% more; $p = .007$); with income added, 2% more (95% CI 8% fewer to 13% more; $p = .688$), while each \$10,000 of median household income was associated with 29% fewer (95% CI 11% to 44% fewer; $p = .004$); with the share of Black residents added, 8% more ($p = .050$); with violent crime added, 12% more ($p = .024$). Limits for poverty: a hidden factor as strong as the share of Black residents, beyond it, could erase the association; one as strong as violent crime would leave it borderline. Method: Cinelli and Hazlett [19], one row per state (log of the 2018 to 2024 rate; $n = 50$), ordinary least squares with conventional standard errors; the weighted row uses weighted least squares with weights equal to the 2023 state population. CI: confidence interval.

Discussion

This study has four main findings. First, mass shootings roughly doubled after 2019 and in 2024 were still well above their 2018 level. Second, higher-poverty states had more mass shootings per million residents: 1.7 times the rate when the 10 highest-poverty and 10 lowest-poverty states are compared, and about 15% more per percentage point of poverty in the main model. That link held without any single state, within regions, and, in the equally

weighted linear sensitivity check, against every factor we measured; in the main model it could not be separated from low income. The two differ because the count model gives large states more weight while the one-row-per-state linear model weights every state equally; when the linear check also weighted states by population, it agreed with the main model, and low income carried at least as much of the link as poverty. We therefore read poverty as one marker of concentrated economic disadvantage, not as a single lever: it could not be separated from low income, it was about half as large once the share of Black residents was considered, and other methods disagreed on which economic measure matters most. Third, state gun-law strength showed no clear link; a within-region link rested on Illinois and a few other states, so state data cannot judge gun laws. Fourth, the lower mass-shooting rates in states with more serious mental illness were too fragile to interpret.

Implications for policy and prevention

Where to focus. Mass shootings were more common in higher-poverty states. Within states, they clustered in a few places: 32% of all incidents happened in just 20 cities or counties. That gives a practical way to set priorities. Prevention and violence-intervention resources could go first to high-poverty states, to the cities and counties where shootings cluster, and to the people and neighborhoods at highest risk there.

Economic conditions belong in the conversation. Of the three state conditions in the main model, poverty was the only one linked to more mass shootings. It was also the finding that held up best, though it could not be separated from low income. Economic and community investment in high-poverty places belongs in the prevention conversation, alongside gun policy, not instead of it. This study cannot say whether such investment lowers mass-shooting rates. It is worth testing in the places where shootings cluster.

Gun laws. This study cannot tell whether state gun laws work. That does not mean gun laws do not matter. A whole state is too large a unit for this question. Most shootings happen in a few places within each state, and one statewide grade cannot capture the laws, enforcement, and gun markets those places face. Answering it needs city or county data, linked to the laws in force at the time.

Mental health. Mental-health care matters for its own reasons. But states with more mental illness did not have more mass shootings. State mental-illness rates are not a guide to where mass shootings happen. This finding is about states, not about individuals.

Associations, not proof. These results show where the burden falls and what goes with it. They do not show that reducing poverty would reduce mass shootings.

Poverty and economic disadvantage

Our results support a link between economic disadvantage and mass-shooting rates, but not a clear answer about which form of disadvantage matters. In the main analysis, income inequality was not a separate link once poverty was considered, unlike in earlier work [2,3]; yet some of the ranking methods placed income and inequality above poverty, poverty could not be separated from low income, and when states were weighted by population, low income rather than poverty carried the link. Merton’s strain theory is about relative deprivation, which fits inequality [4]. A state-wide inequality index is, however, a poor stand-in for the neighborhood-scale gaps that strain theory describes, so state data cannot test the theory well. The four-or-more-shot definition captures much everyday community gun violence (only 5% of the incidents here had four or more people killed), and firearm injury in U.S. cities concentrates in the neighborhoods shaped by discriminatory lending and disinvestment [23,24]. This is also how we read the strong link with the share of Black residents: as a marker of segregation and concentrated disadvantage, not of race as a cause. Measures of segregation and concentrated disadvantage at the neighborhood scale, linked to where incidents happen, would be the right next test.

Gun laws

Our gun-law result is an absence of evidence, not evidence of absence. Earlier studies are mixed and depend on the definition used: studies of fatal (four-or-more-killed) shootings found more permissive state gun laws linked to higher rates [5], whereas an analysis using the broader four-or-more-shot definition, as here, found no link with state gun-law grades [6]. Our main result agrees with the latter. The positive within-region estimate rested largely on Illinois, which has the strongest gun-law grade in the Midwest and more incidents than any other state (375, 74% of them in Chicago); a comparison like this may say more about one city than about state law. The gun-law measure is also coarse: 21 of the 50 states share the lowest grade. Household gun ownership, which tracks gun-law strength closely and was measured in 2016, could not be separated from it. Finally, states with more shootings may later strengthen their laws, which would make stronger laws appear linked to more shootings. Recent evidence argues against one route for this: mass shootings in or near a state legislator’s district did not, on average, change how that legislator voted on gun bills [25]. A gun-law measure that changes over time, at a finer geographic scale, would be the stronger test.

Mental illness

The mental-illness result is about states, not the people who commit mass shootings. Research on individuals addresses a different question: a review concluded that serious mental illness is neither necessary nor sufficient for mass violence [26], and in a study of 172 mass shooters, psychosis played no role in most cases [27]. The state estimates are also imprecise: they are survey-based estimates that pool two years [12,28], and their published 95% intervals (typically 2.6 percentage points wide) are wider than the typical difference between states. What these data do show is that states with more mental illness did not have more four-or-more-shot mass shootings, a definition that selects different incidents from fatality-based ones [8]. The lower rates in such states in the main model were fragile: a very weak hidden factor would make them unclear, and adding the share of Black residents erased them. We do not read them as meaningful.

Strengths and limitations

Strengths include seven years of data on all 50 states, state characteristics taken directly from documented official files, one script that produces every reported number, a poverty finding that held up when any state was left out and when states were compared within regions, a check of how strong a hidden factor would have to be to change each finding, and full reporting of the analyses that pointed another way (Additional file 4).

The study has five main limitations.

1. It is observational and compares states. The links it finds are associations, not causes, and they say nothing about individuals. A hidden factor as strong as the share of Black residents, beyond it, could erase the poverty link.
2. The state is a coarse unit. Mass shootings concentrate in a few cities, and 50 states is a small number of comparisons.
3. The four-or-more-shot definition mainly captures community gun violence; only 5% of incidents had four or more people killed. Results may differ for fatal-only or public mass shootings.
4. Each state characteristic was measured at one time (mostly 2022 or 2023), while the shootings span 2018 to 2024. The study can document the surge but cannot explain it, and it cannot show whether changes in poverty or in laws came before changes in shootings.
5. Poverty, low income, and the share of Black residents overlap across states, and with 50 states the study cannot separate them or say which matters; different analysis methods

also disagreed, and whether poverty stayed linked with income added depended on whether large states counted more. The share of Black residents was itself strongly linked to mass-shooting rates. We read it as a marker of residential segregation and concentrated disadvantage, not as a cause, and it says nothing about individuals.

Conclusions

Mass shootings in the United States roughly doubled after 2019 and stayed high through 2024. They were more common in higher-poverty states, and they clustered in a few cities and counties. Poverty, low income, and segregation travel together, and this study cannot tell which matters most. Of the three conditions we tested, poverty was the one linked to more mass shootings, and that link held up best; but it could not be separated from low income, which carried at least as much of the link when large states counted more. Most incidents counted here were not mass killings, so results may differ for fatal or public mass shootings. These findings are associations. They do not prove that reducing poverty would reduce mass shootings.

The takeaway is simple. Where poverty and disadvantage concentrate, mass shootings concentrate too. Prevention resources belong in those places, and economic investment there is worth testing, alongside gun policy. State data cannot settle whether gun laws work; that question needs local data. And state mental-illness rates are not a guide to where mass shootings happen.

Declarations

Ethics approval and consent to participate

Not applicable. The study used only publicly available, de-identified aggregate state-level data; no human subjects were involved.

Consent for publication

Not applicable.

Availability of data and materials

The analysis dataset (Additional file 1), its data dictionary with the source, table, and URL of every variable (Additional file 2), the racial and ethnic composition covariates with their

dictionary (Additional file 3), and the code that rebuilds the datasets from the official files and produces every reported number will be deposited at the Open Science Framework; until then they are available from the author on request. The deposited outcome consists of derived counts of mass-shooting incidents per state and year, not Gun Violence Archive incident records; incident-level data are available from the Gun Violence Archive (<https://www.gunviolencearchive.org/>), which retains copyright in them.

Competing interests

The author declares no competing interests.

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Additional files

Additional file 1: Analysis dataset: state-level mass-shooting counts by year (2018 to 2024) and all state characteristics (Dataset_v3_official.csv). (CSV)

Additional file 2: Data dictionary: definition, unit, source, table or field, and URL for every variable (DataDictionary_v3_official.csv). (CSV)

Additional file 3: Racial and ethnic composition covariates, ACS 2023 1-year tables B02001 and B03002 (covariates_race_v3_1.csv), with their data dictionary (DataDictionary_race_v3_1.csv). (CSV)

Additional file 4: Tables S1 to S5: the state characteristics used in this paper and their data sources (S1, S2), and every other analysis we ran, in plain words: methods that rank state characteristics (S3), other versions of the main model (S4), and each characteristic on its own (S5). (PDF; appended below)

Additional file 4: State characteristics, data sources, and other analyses

This file accompanies “Economic Disadvantage and Mass Shootings Across U.S. States, 2018 to 2024: A State-Level Panel Analysis”. Reference numbers refer to the main-text reference list.

Table S1: State characteristics used in this paper (50 states)

Characteristic	Year	Mean	SD	Min	Max
Mass shootings per million residents per year	2018 to 2024	1.4	1.2	0.0	5.6
Poverty rate (%)	2023	12.2	2.6	7.2	18.9
Gun-law strength (0 = F to 4 = A)	2023	1.56	1.61	0.00	4.00
Serious mental illness, adults (%)	2022 to 2023	6.36	1.04	4.64	9.34
Any mental illness, adults (%)	2022 to 2023	24.08	2.48	19.69	30.16
Median household income (\$)	2023	76,976	12,262	54,203	99,858
Gini index of income inequality	2023	0.466	0.017	0.423	0.516
Black or African American residents (%)	2023	10.1	9.0	0.4	35.6
Hispanic or Latino residents (%)	2023	13.3	10.3	2.1	48.6
Households with a gun (%)	2016	39.7	14.6	8.9	65.0
Violent crimes per 100,000 residents	2023	355.2	146.1	103.8	746.9
Urban population (%)	2020	72.4	14.8	35.1	94.2

Unweighted summaries across the 50 states (District of Columbia excluded). Mass shootings: each state’s 2018 to 2024 total divided by 7 and by its July 1, 2023 population. SD: standard deviation. Sources in Table S2 and Additional files 2 and 3.

Table S2: Data sources

Variable	Source	Table or field
Mass shootings by state and year	Gun Violence Archive, incident-level copy dxzys/Gun-Violence-Data, commit 3d2c2b8 [1,9]	Incidents with four or more shot, counted by incident ID, state, and calendar year
Population	U.S. Census Bureau, Population Estimates Program, Vintage 2024 [13]	July 1, 2023 estimate (POPESTIMATE2023)
Census region	Same file [13]	REGION
Poverty rate	ACS 2023 1-year [10]	B17001 (below poverty level / poverty status determined)
Median household income	ACS 2023 1-year [10]	B19013
Gini index	ACS 2023 1-year [10]	B19083
Black or African American residents	ACS 2023 1-year [10]; Additional file 3	B02001 (Black or African American alone / total)
Hispanic or Latino residents	ACS 2023 1-year [10]; Additional file 3	B03002 (Hispanic or Latino / total)
Gun-law strength	Giffords Law Center 2023 scorecard, Internet Archive copy of 1 April 2024 [11]	Letter grade per state, converted to 0 to 4 (Methods)
Serious mental illness	SAMHSA, 2022-2023 NSDUH state prevalence estimates [12]	Table 34, adults aged 18 or older
Any mental illness	Same [12]	Table 33, adults aged 18 or older
Households with a gun	RAND state-level estimates of household firearm ownership [14]	Household firearm ownership rate, 2016 (latest year estimated)
Violent crime	FBI Uniform Crime Reporting, estimated crimes [15]	Violent crimes per 100,000 residents, 2023
Urban population	2020 Census urban and rural state file [16]	Percent of population in urban areas

Additional file 2 gives the exact field and download address of every variable in the shared dataset, including the further characteristics used only by the ranking methods in Table S3.

Other analyses

The main paper reports one main model and three checks. This section gives every other analysis we ran on these data, including those that point another way. Unless stated otherwise, results are the percent change in the mass-shooting rate per million residents for one natural unit, as in the main text, with its 95% confidence interval (CI) and p value, and the year accounted for. “Clearly associated” means the 95% CI excludes no change.

Methods that rank state characteristics (Table S3). These methods ask which of 14 state characteristics best predict each state’s 2018 to 2024 mass-shooting rate, using one row per state. With only 50 states they are a rough check, not a test, and they can disagree when characteristics move together, as poverty, income, and inequality do.

- *Random forest* [21] grows many decision trees, each splitting states into groups with

similar rates, and ranks each characteristic by how much worse the predictions get without it (average over 50 runs).

- *Gradient boosting* [22] grows small trees one after another, each correcting the errors of the last, and ranks characteristics by how much they improve the predictions.
- *LASSO* [20] is a regression that drops characteristics that add little. We report how often each characteristic was kept over 200 repeated runs, in two versions: one on each state’s count of incidents (accounting for population), one on its rate.

Table S3: Which characteristics the ranking methods placed highest

Method	Poverty	Median household income	Income inequality (Gini)	Gun-law strength	Placed first
Random forest (rank of 14)	1	4	3	13	Poverty
Gradient boosting (rank of 14)	10	1	2	14	Median household income
LASSO, count version (share of runs kept)	54%	94%	0%	<1%	Single-person households (100%), then income
LASSO, rate version (share of runs kept)	100%	1%	100%	1%	Poverty, inequality, violent crime, mental-health providers, any mental illness (each 100%)

Rank 1 is the strongest predictor. The 14 characteristics: poverty, median household income, Gini index, gun-law strength, households with a gun, any mental illness (used in place of serious mental illness, with which it is closely correlated, $r = 0.86$), mental-health providers per 100,000, substance use disorder, violent crime, population density, percent urban, percent aged 18 to 29, percent single-person households, and alcohol consumption (sources in Additional file 2). The rankings say which characteristics predict best, not in which direction or why. LASSO choices were also unstable: with a stricter penalty the count version kept no characteristic at all, and when the 50 states were resampled 200 times, each characteristic was kept in between 30% and 94% of runs.

Other versions of the main model (Table S4). Each row changes one thing about the main model. Most leave the three main results as they are: poverty associated, gun-law strength not clearly associated, and serious mental illness not associated with more mass shootings. The exceptions are these. With income added, poverty was no longer clearly associated (main text). Within Census regions, stronger gun laws were associated with more mass shootings, including when racial composition or percent urban was added; this rested on a few states (main text). Fewer mass shootings in states with more serious mental illness appeared in several versions but not within regions or with racial composition added. In place of poverty, median household income was clearly associated (higher income, fewer mass shootings), while income inequality was not.

Table S4: Other versions of the main model

Version	Poverty, per percentage point	Gun-law strength, per full letter grade Change in rate, % (95% CI); p	Serious mental illness, per percentage point
Main model (for comparison)	15 (6 to 26); .003	4 (−10 to 20); .595	−18 (−31 to −2); .032
Income inequality in place of poverty	Gini per 0.01: 17 (−2 to 39); .076	−10 (−22 to 4); .145	−17 (−32 to 2); .076
Median household income in place of poverty	Income per \$10,000: −32 (−46 to −15); .002	14 (−3 to 33); .109	−20 (−33 to −5); .013
Households with a gun and violent crime added	11 (1 to 22); .033	14 (−2 to 33); .077	−27 (−39 to −13); .002
Median household income added	8 (−6 to 24); .241	12 (−6 to 34); .183	−18 (−32 to −2); .030
Share of Black residents added	8 (1 to 15); .026	8 (−6 to 24); .253	3 (−14 to 22); .739
Shares of Black and Hispanic residents added	9 (2 to 16); .017	9 (−5 to 24); .206	−1 (−19 to 21); .939
Comparing within Census regions	16 (6 to 26); .003	19 (2 to 38); .025	−6 (−23 to 14); .468
with Black and Hispanic shares added	11 (4 to 18); .006	15 (1 to 31); .036	16 (−5 to 42); .145
with percent urban added	16 (7 to 27); .003	17 (1 to 37); .042	−5 (−23 to 17); .614
Percent urban added	15 (5 to 26); .005	4 (−11 to 22); .578	−18 (−31 to −2); .031
Each year's own population instead of 2023	15 (6 to 26); .004	3 (−11 to 20); .635	−18 (−31 to −2); .031
One row per state (2018 to 2024 totals)	19 (10 to 29); < .001	8 (−6 to 25); .271	−20 (−33 to −5); .016
Each state with its own baseline rate	19 (10 to 28); < .001	6 (−7 to 21); .377	−21 (−34 to −5); .011
2018 to 2019 only	14 (2 to 27); .024	2 (−16 to 22); .863	−20 (−37 to 1); .062
2020 to 2024 only	16 (6 to 26); .003	5 (−9 to 20); .512	−17 (−30 to −2); .032
Without the small-sample correction	15 (7 to 25); < .001	4 (−9 to 19); .573	−18 (−30 to −4); .013

Each cell: percent change in the mass-shooting rate per million residents for one unit, with its 95% confidence interval (CI), then its p value. In the version with households with a gun and violent crime added, gun ownership was associated with 24% more per 10 percentage points of households (95% CI 4% to 46% more; $p = .017$) and violent crime with 13% more per 100 crimes per 100,000 residents (95% CI 2% fewer to 30% more; $p = .090$); in that version, fewer mass shootings in states with more serious mental illness held when any one state was left out (50 of 50 refits). The difference in the poverty association between 2018 to 2019 and 2020 to 2024 was not clear ($p = .793$). Standard errors: CR2 cluster-robust by state with Satterthwaite degrees of freedom, as in the main model. One row per state: each state is its own cluster, so CR2 reduces to the HC2 heteroskedasticity-robust standard error. Each state with its own baseline rate: a model with a random intercept for each state, with model-based (Wald) standard errors. Without the small-sample correction: CR1 instead of CR2 standard errors, which gives narrower intervals.

Each characteristic on its own (Table S5). Each row is a model with one characteristic and the year. These show the raw pattern before anything else is held equal.

Table S5: Each characteristic on its own

Characteristic, per unit	IRR per unit (95% CI)	Change in rate, % (95% CI)	<i>p</i>
Poverty rate, per percentage point	1.14 (1.05 to 1.23)	14 (5 to 23)	.003
Median household income, per \$10,000	0.80 (0.66 to 0.96)	−20 (−34 to −4)	.022
Gini index, per 0.01	1.23 (1.05 to 1.43)	23 (5 to 43)	.015
Gun-law strength, per full letter grade	0.95 (0.82 to 1.10)	−5 (−18 to 10)	.479
Households with a gun, per 10 percentage points	1.09 (0.91 to 1.29)	9 (−9 to 29)	.332
Serious mental illness, per percentage point	0.80 (0.65 to 0.97)	−20 (−35 to −3)	.026
Any mental illness, per percentage point	0.89 (0.82 to 0.97)	−11 (−18 to −3)	.008
Violent crime, per 100 crimes per 100,000 residents	1.18 (0.99 to 1.40)	18 (−1 to 40)	.063
Black residents, per 10 percentage points	1.69 (1.46 to 1.95)	69 (46 to 95)	< .001
Hispanic residents, per 10 percentage points	0.88 (0.75 to 1.02)	−12 (−25 to 2)	.088

Each row: negative-binomial model with that characteristic, calendar-year fixed effects, a population offset, and CR2 standard errors by state with Satterthwaite degrees of freedom, as in the main model. IRR: incidence rate ratio; CI: confidence interval. The share of Black residents is read as a marker of segregation and concentrated disadvantage, not as a cause (main text).